PAYMANSHU SHARMA

+91-9810032357 | paymanshu@gmail.com | Bangalore, KA, India
linkedin.com/in/paymanshus | github.com/Paymanshus | paymanshus.github.io | @paymanshu

EXPERIENCE

Maersk
April 2025 - Present
Machine Learning Engineer
Bangalore, India

- Building Shipment CoCaptain for financial performance management: an agentic insights copilot over shipment and financial data, cutting manual analysis from 3-4 weeks per request to minutes
- Won sponsorship by pitching to area CFOs (North America, East Med), finance business partners, and BI analysts, with recurring CTIO demos; selected as 1 of ~20 teams org-wide to present at Copenhagen HQ
- Ran a Copenhagen HQ workshop with engineering leadership on building an AI-first org with forward-deployed teams; helped define the team's FDE and AI/ML hiring blueprint; advised on right-sizing AI over default multi-agent complexity
- Patent filed: Method and System for Optimising Computational Resource Usage during Query Processing (DKPTO, Apr 2026) for context management in agentic AI workflows: precise state architecting with injectable domain rules, definitions, and formulae for grounding and safety, cutting per agent/node context by up to 60% vs full-state propagation
- Shipped the product end-to-end on legacy financial data: LangGraph orchestrator-evaluator pipeline (planning, structured outputs, parallel execution, recursive error handling, LLM-as-a-judge, citation grounding), React frontend (async polling, citation tooltips, LLM-generated Plotly charts), and Azure PostgreSQL with RBAC
- Delivered an executive KPI dashboard with AI-generated widgets in under 2 days, challenging Power BI/Excel workflows; spec-driven development from a design.md built on Maersk's design system, with a Redis-cached multi-source API
- Scaling payments automation across 12+ countries: document-AI pipelines (Pay to Release, Payapps) that read payment proofs and remittance emails with Azure Document Intelligence and LLMs, check them against SAP, and auto-approve or post; a few thousand documents/day worth ~$1.3M/day, targeting ~21 FTEs of manual work saved
- Made new-country launches a config change: per-country prompt configs, benchmarks, and regression evals gating every launch without regressing live markets; iterative evals cut human-in-the-loop review toward ~80% zero-touch
- Owned ingestion reliability: lost submissions cut from 30-60/day to under 5 (queue-first acceptance, hash-based idempotent dedup, retries); SLOs for 99.9% no-loss and 30-min processing, with SLA-priority ordering under SAP rate limits
- Used SLO-driven observability (Prometheus, Grafana, Loki, drift alerts) to catch automation-rate dips, error spikes during volume surges, and silent failures early; dashboards-as-code built for support handover

Hike (Rush Gaming Universe)
December 2024 - April 2025
Machine Learning Engineer
Remote (Bangalore, India)

- Built the AI outreach manager end to end: a real-time multilingual voice bot for Hinglish speakers in Tier-2 India, underserved by generic voice AI, built for high-trust engagement over cold ad-style outreach
- Cut voice bot reply latency by ~700ms (30%) in the STT-LLM-TTS pipeline using filler responses and interim results; queue-based scheduler for async call requests at scale via SIP trunking
- Set up a configurable template for deploying customized bot agents with modifiable guardrails and speech guidelines, enabling personalized user engagement and offer targeting

Mphasis NEXT Labs (R&D Labs)
December 2021 - December 2024
Lead Data Scientist (Prev. Data Science Intern)
Bangalore, India

- Translated client requirements into solution designs across enterprise RFP/RFI cycles, building proof-of-concepts on client data and pitching to mixed business and technical audiences
- Spearheaded information extraction in DeepInsights, a document intelligence suite for enterprise clients: fine-tuned open-source LLMs with QLoRA on low-resource compute, optimized inference throughput with vLLM, and outperformed Azure Document Intelligence and Textract on client datasets of only ~1K documents
- Implemented a novel variant of DeBERTa's disentangled attention that adds document positional information, for a data extraction model trained from scratch, with a task-specific custom evaluation metric
- Designed a multi-agent conversational architecture with an LLM-powered recommendation system (two-tower model, zero-shot retrieval, long-context user and product insights) with explainable results
- Led an intelligent document processing app for mobile devices with extraction and RAG-based QA, optimizing models with ONNX runtime and quantization; directed a scalable SaaS product on AWS (ECS/Fargate microservices, Lambda + SQS driven GPU autoscaling)
- Applied research across a multi-modal fusion transformer for audio-video-text retrieval, federated learning on edge devices (Flower + AWS Greengrass), LIME/SHAP explainability, and autoencoder-based data imputation (+8% downstream accuracy)

MetFlux Research (IIT Bombay) June 2021 - September 2021
Computer Vision and Signal Processing Research Intern Remote (Mumbai, India)

- Built the PPG signal processing and validation pipeline for a low-cost at-home health-check app (HR, RR, SpO2), estimating vitals from face and fingertip video; improved MAE by 55% (Prof. K. V. Venkatesh)

Amicus.ai February 2021 - June 2021
Founding Team Member and Data Scientist Remote (Mumbai, India)

- Built v1 of a legaltech product bringing NLP to legal case analysis: BERT entity recognition and rule-based extraction on unstructured legal text from 100,000+ case files, with Dash/Plotly visual analytics

Awarathon July 2019 - December 2019
AI Intern Mumbai, India

- Built speech and vision scoring (cadence, tone, gaze, gestures, transcription) for interview screening on 200+ clips

PROJECTS

Multilingual AAC Communication App | React Native, Expo, ElevenLabs, Cartesia

Building an assistive communication (AAC) app with native Hindi-English code-switching, personalized next-token phrase prediction and a three-tier latency architecture for TTS (pre-rendered cache, on-demand cached, streaming) balancing instant response against synthesis cost

CereBro: Auditable Opinion Research Engine | Next.js, React, Supabase, Vercel

Built a deep-research opinion mining engine across Reddit, HN, X, and the web: the LLM extracts and clusters claims, deterministic weighting with snippet-level source traceability enables instant zero-LLM-call re-scoring on source re-weighting

Agent Skills | Claude Code, Agent Skills spec, npx | github.com/Paymanshus/skills

Published model-agnostic agent skills for AI-assisted coding built around cost-minimization: tiered task routing, one-time codebase specs over repeated exploration, and planning workflows compiling into parallel multi-session execution packages

AudioAnalytics: Emotion Detection Package | Python, TensorFlow, Librosa, PyPI

Published an open-source PyPI package that extracts audio features and predicts emotion with a 1D CNN model

Gaze Estimation and Saliency Prediction Pipeline to Study Visual Attention for E-Commerce Websites
(Undergraduate Thesis)

HONORS & AWARDS

LLM Prompt Recovery Challenge: Bronze Medal [Kaggle Competition]

Top 9% of 2200 participants as a solo entrant, working on prompt jailbreaking and evaluation

Bajaj Finserv Hackathon: National-level Runners-Up [Search and Recommendation Engine]

2nd overall, 1st in category among 3000+ teams; built a BERT-embedding recommendation engine for Finserv's search

ACM Mumbai Hackathon (Region-level) and SIH Internal Hackathon (College-level): Winners [Prototyping, pitching]

Ideated, built, and pitched Entrenet (a Flutter network for founders, investors, job seekers) and an ML crop-detection app

SKILLS

AI/ML: LLMs, Agentic Systems, LangGraph/LangChain, MCP, RAG & GraphRAG, Evals & LLM-as-a-Judge, LangSmith/Langfuse, Context Engineering, Fine-tuning (LoRA/QLoRA), vLLM, ONNX, PyTorch, Transformers, NLP, Document AI, Voice AI, Recommendation Systems

Engineering: Python, SQL, TypeScript, React, Next.js, FastAPI, PostgreSQL, Redis, Azure Service Bus, Docker, Microservices, CI/CD, Prometheus, Grafana, Loki, AWS, Azure, GCP

Product & Delivery: Stakeholder Discovery, Solution Scoping, PoC-to-Production, Executive Demos & Pitching, Spec-Driven Development, Rapid Prototyping

EDUCATION

MPSTME, Narsee Monjee Institute of Management Studies, Mumbai 2018 - 2022
Bachelor of Technology in Data Science - CGPA: 3.82/4.0 Mumbai, India

Delhi Public School, Gurgaon 2018
President, NODE (Computer Club) Gurgaon (NCR), India

- Managed a 50+ member club across grades, organized district-level inter-school events

EXTRACURRICULARS

Rotaract Club of Bombay Airport July 2019 - July 2021
Led the Open School program reaching 150+ underprivileged students, with national news coverage